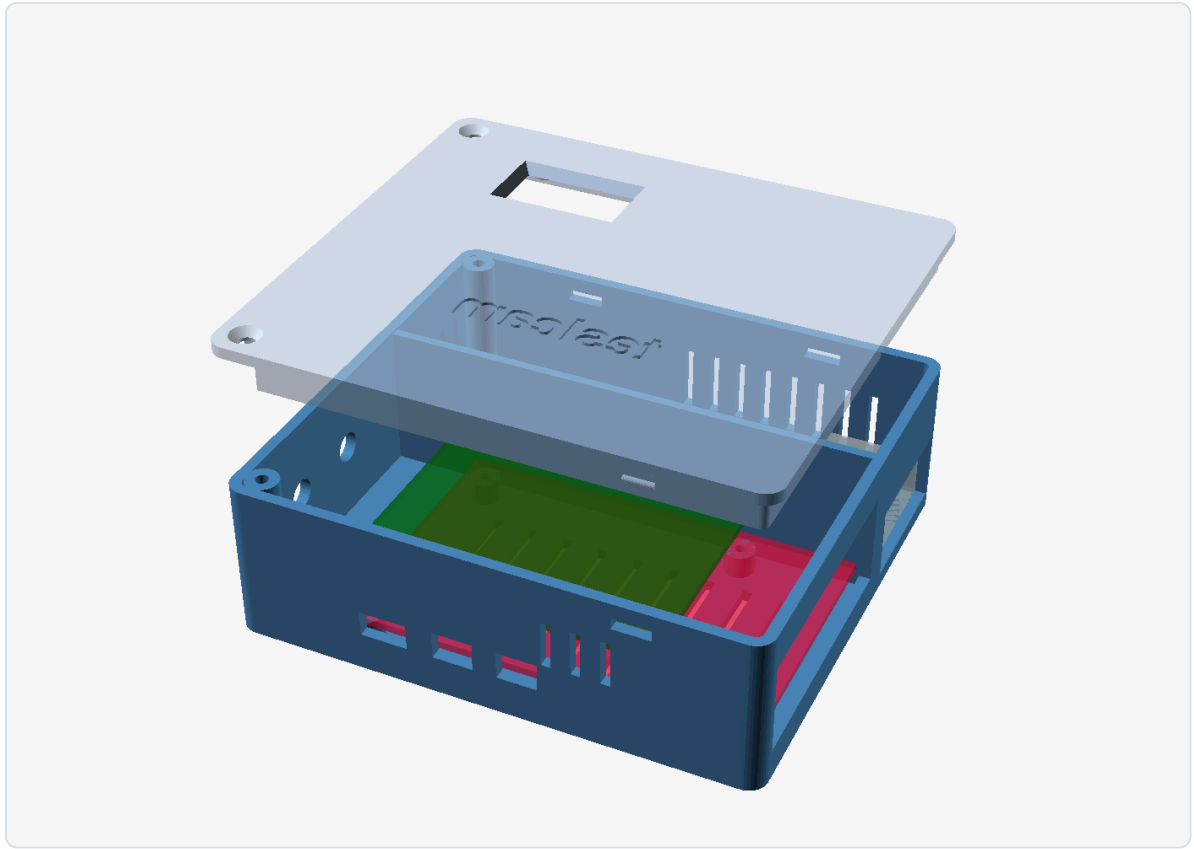


teslcam in-car unit — case assembly DRAFT v0.1

3D-printed enclosure for Raspberry Pi 4B · Waveshare SIM7600 LTE HAT · M.2 USB SSD · 0.96" OLED

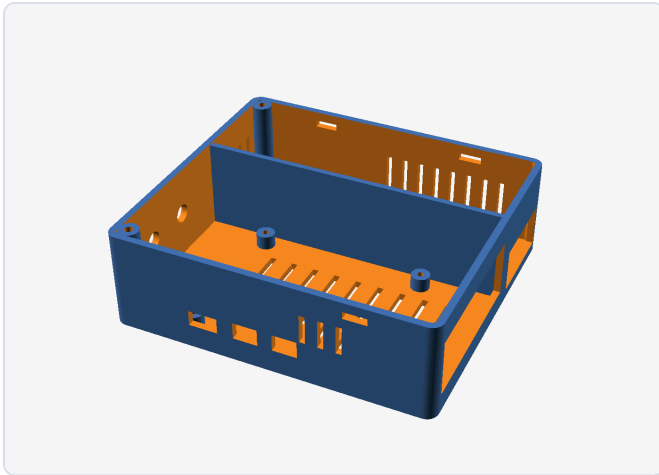


Exploded view — lid (with OLED window and rear screw holes) above the two-compartment base. Ghosted: Pi 4 (red) with HAT above it (green), SSD enclosure (silver).

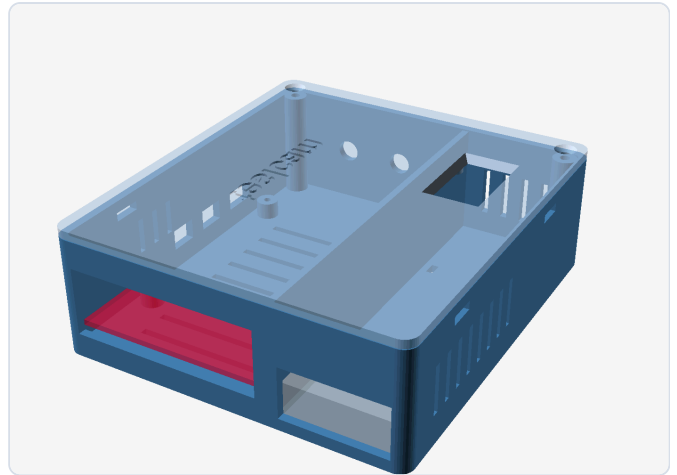
Early draft. All dimensions are parametric in `teslcam-case.scad` but were chosen from datasheet-typical values, not measured hardware. Dry-fit every board before final assembly; expect a v0.2.

Layout

One tray, two compartments. The Pi + HAT stack sits on four standoffs in the wide compartment; the SSD enclosure drops into the narrow bay across the divider. All connections face the same end wall: the Pi's USB/Ethernet ports sit behind the large opening, and both USB cables (HAT and SSD) loop externally back into the Pi's ports. Antenna pigtailed exit through two SMA holes in the rear wall from the dead zone behind the Pi. The OLED screws to the lid, facing out through its window over the SSD bay.



Base: Pi standoffs (right), SSD bay (left of divider), SMA holes in the rear wall, floor vents under the Pi, rear lid-screw pillars.



Port end, closed: Pi port bay (red) and SSD cable mouth (silver) share the end wall.

Printing

Part	File	Orientation	Settings
Base	base.stl	As modeled — floor on bed	0.2 mm layers, 3 perimeters, no supports. PETG or ASA — a parked car gets hot enough to warp PLA.
Lid	lid.stl	As modeled — flat face on bed, lip up	

Footprint $\approx 111 \times 100 \times 37$ mm. The lid flips over onto the base after printing; the engraved label lands on the outside.

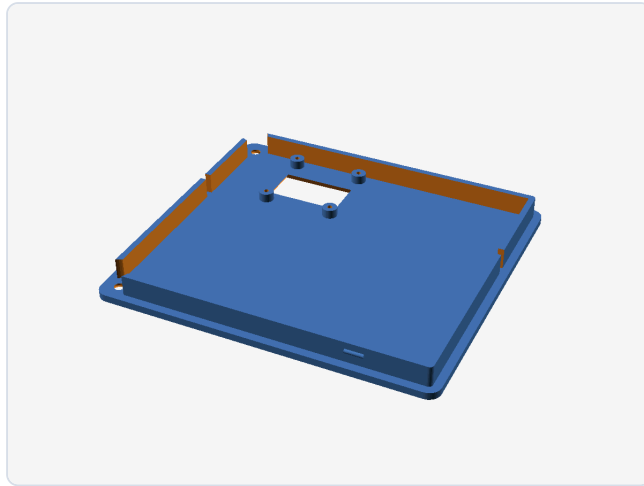
Bill of materials

Qty	Item	Used for
1	Raspberry Pi 4B	
1	Waveshare SIM7600 4G HAT + nano-SIM	LTE uplink
1	M.2 SSD in USB 3 enclosure ($\leq 100 \times 31 \times 10$ mm)	Clip storage
1	SSD1306 0.96" OLED, I ² C	Status display
4	M2.5 $\times$ 6 screws	Pi $\rightarrow$ case standoffs
4	M2.5 11 mm male–female standoffs + top screws	Pi $\rightarrow$ HAT stack
2	M3 $\times$ 8 self-tapping screws	Lid, rear corners
4	M2 $\times$ 4 self-tapping screws	OLED $\rightarrow$ lid posts
2	SMA bulkhead pigtails (IPX/U.FL $\rightarrow$ SMA, ≥ 10 cm)	LTE MAIN + GPS
2	Antennas (SMA)	
1	USB-A $\rightarrow$ micro-B cable, ~ 15 cm	Pi USB 2 $\rightarrow$ HAT
1	USB-A $\rightarrow$ enclosure cable, ~ 15 cm	Pi USB 3 (blue) $\rightarrow$ SSD
4	Female–female jumper wires	OLED I ² C
2	Zip ties (or foam pad)	SSD retention

Assembly

1. **SIM first.** Insert the SIM into the HAT now — the slot is hard to reach once the stack is boxed.
2. **SMA bulkheads.** Fit both pigtails through the rear-wall holes, nut on the outside. Coil the slack in the dead zone behind the Pi position.
3. **Pi in.** Seat the Pi on the four standoffs, ports toward the big end opening; fasten with four M2.5 $\times$ 6 screws. The microSD stays reachable from inside while the lid is off.
4. **OLED wiring before the HAT.** Attach the four jumpers to the Pi header — 3V3 (pin 1), SDA (3), SCL (5), GND (9) — and route them over the divider into the SSD bay. The HAT covers the header next, so this order matters.
5. **HAT stack.** Thread the 11 mm standoffs into the Pi mounting holes, press the HAT onto the GPIO header, and secure with the top screws. Click the pigtails' IPX ends onto the HAT's MAIN and GPS connectors.
6. **SSD.** Drop the enclosure into the bay, cable toward the mouth, and strap it down with zip ties through the floor slots (or wedge with a foam pad).
7. **External USB loops.** Outside the port wall, connect: HAT micro-B $\rightarrow$ Pi USB 2 port; SSD cable $\rightarrow$ Pi USB 3 (blue) port.
8. **OLED to lid.** Screw the display onto the four posts (M2), glass facing the window, and connect the jumpers.

9. **Close.** Angle the lid in, press until the three snap bumps click into their windows, then drive the two rear M3 screws. Power arrives by USB-C through its side cutout.



Lid as printed: perimeter lip with snap bumps, four OLED posts around the window, countersunk rear screw holes.

Known gaps in v0.1

- SSD1306 hole pitch varies by vendor — `oled_hole_pitch` assumes 23.5 mm; measure yours. Tune `oled_win_dy` for the glass offset after a test print.
- No strain relief on the external USB loop yet; a printed clip is planned.
- If the HAT's own SMA jacks foul the wall, raise `pi_clear` or notch the wall.
- Passive cooling only; v0.2 adds an optional 30 mm fan boss if the Pi throttles.
- No glovebox mounting ears/velcro base yet.